

CSCI 5541 NLP: HW 3

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1 Introduction

This report covers the implementation, results, analysis, and future work for CSCI 5541 HW 3: Fine-tuning LLMs for Text Classification.

2 Implementation

2.1 Dataset Selection

For this implementation, the DeepfakeTextDetect dataset from Hugging Face was selected. The DeepfakeTextDetect dataset includes both human-written texts and texts generated by different large language models (LLMs). The dataset contains diverse examples to capture stylistic and linguistic differences between human and machine-generated texts.

The task is a binary classification problem, where the model predicts whether a given input text was written by a human (label 0) or generated by AI (label 1). The dataset was split into training, validation, and testing sets. The training set was used to fine-tune the pretrained model. The validation set was used to monitor performance during training and select the best model checkpoint. The test set was used for final evaluation.

2.2 DistilBERT Model Implementation

The pretrained DistilBERT model from Hugging Face was selected as the baseline model for fine-tuning. The distilbert-base-uncased checkpoint was used and a sequence classification head with two labels was added for binary classification. The Hugging Face Trainer API was used to handle the training loop, evaluation, and metric computation. Text inputs were tokenized using the corresponding DistilBERT tokenizer with truncation enabled to handle long sequences. The batch size was set to 16 (the default was 8) for faster training. This does not affect experimental validity because batch size is not a controlled variable.

The validation size was set to 2000 for faster training. This was done to reduce per-epoch evaluation time while maintaining sufficient sample size. Dynamic padding with DataCollatorWithPadding was applied during batching using a data collator to improve training efficiency.

The following experimental variables were explored: number of epochs, training data size, and learning rates. Epoch values of 1 through 10 were trained with a fixed training size of 800 and fixed learning rate of $2e-5$. Training data sizes of 50, 100, 500, and 1000 were trained with 4 epochs and a fixed learning rate of $2e-5$. Learning rate values of $1e-5$, $2e-5$, and $3e-5$ were trained with 4 epochs and a fixed training size of 2000.

3 Results and Analysis

3.1 Test Performance and Plots

The epoch, training size, and learning rate plots are depicted in Figures 1, 2, and 3 respectively. In Figure 1, validation accuracy peaks at a value of about 0.525 at 4 epochs. In Figure 2, validation accuracy peaks at a value of about 0.5667 at a training size of 1000. In Figure 3, validation accuracy is flat at a value of about 0.495 at all learning rate values of $1e-5$, $2e-5$, and $3e-5$.

3.2 Analysis

3.2.1 Number of Epochs

Figure 1 shows the relationship between the number of training epochs and validation accuracy. Performance is relatively low at small epoch values (1–2), indicating underfitting where the model has not yet learned meaningful patterns from the data. Accuracy improves as training continues and reaches its peak at approximately 4 epochs. Beyond this point, performance gradually declines between epochs 6–10, suggesting the model begins to overfit to the training data rather than generalizing to unseen examples.

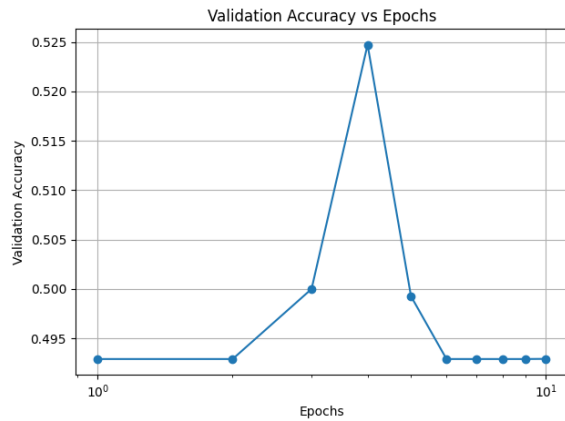


Figure 1: Validation accuracy plotted against epochs (ranging from 1 to 10).

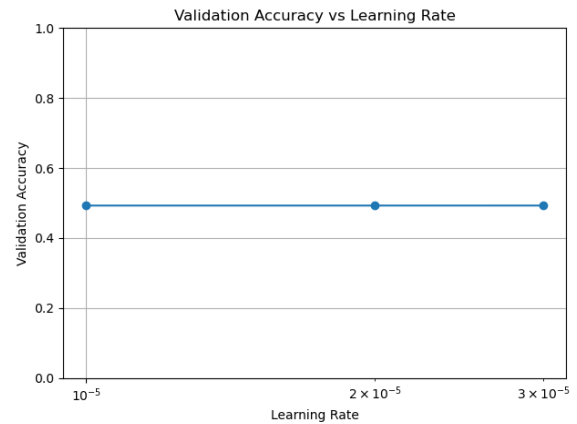


Figure 3: Validation accuracy plotted against learning rate (1e-5, 2e-5, and 3e-5).

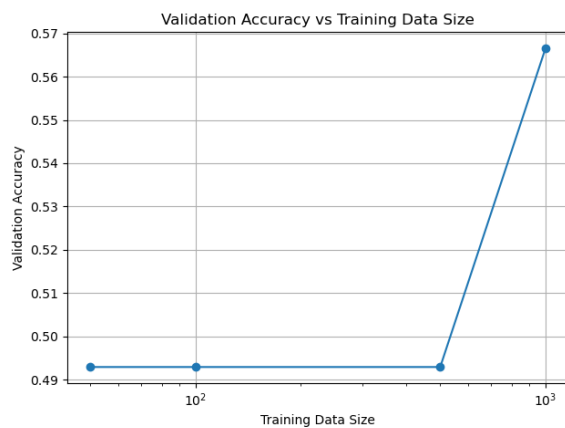


Figure 2: Validation accuracy plotted against training size (50, 100, 500, and 1000).

Overall, mid-range epoch values (3–5) appear to provide the best balance between sufficient training and generalization, with the highest validation accuracy occurring at 4 epochs. These results illustrate that simply increasing training duration does not necessarily lead to better performance.

3.2.2 Training Data Size

Figure 2 illustrates how validation accuracy changes as the amount of training data increases. For smaller training sizes (50, 100, and 500 samples), accuracy remains close to 0.49, indicating that the model struggles to learn meaningful distinctions between human-written and AI-generated text when limited data is available.

A noticeable improvement occurs when the training size increases to 1000 samples, where validation accuracy rises to approximately 0.5667. This suggests that larger training datasets provide the model with more stylistic and linguistic variation,

allowing it to better capture patterns that differentiate human and machine-generated text.

It is also important to note that the validation set was fixed at 2000 samples for efficiency, which results in an imbalance when very small training subsets are used. Because the dataset split does not strictly follow the standard 8:1:1 ratio, the limited training data may contribute to underfitting and relatively low overall accuracy.

3.2.3 Learning Rate

Figure 3 presents validation accuracy across three learning rate values: 1e-5, 2e-5, and 3e-5. Performance remains nearly identical for all tested values, with validation accuracy remaining around 0.495.

This flat trend suggests that the model is relatively insensitive to small learning rate variations within this range. Transformer-based models such as BERT and DistilBERT are known to train stably when fine-tuned within the commonly recommended range of 1e-5 to 5e-5. Since all tested learning rates fall within this interval, they likely reside within a stable optimization region that produces similar performance outcomes.

3.3 Best Combination of Parameters

After exploring the experimental variables, the best-performing configuration among the tested settings used 4 training epochs, a training size of 1000 samples, and a learning rate of 2e-5. This configuration achieved a validation accuracy of 0.4929 and a test accuracy of 0.4942.

Training for 4 epochs appeared to provide a reasonable balance between underfitting and overfitting. When the number of epochs was very small, the model likely did not have sufficient time to

adapt its pretrained representations to the human-versus-AI text detection task. In contrast, larger epoch values showed declining validation accuracy, suggesting the model may begin to overfit the training data. This behavior is consistent with common fine-tuning practices for transformer models, where BERT-style architectures often achieve strong performance when trained for only a few epochs (Devlin et al., 2019). In this experiment, 4 epochs produced the highest validation accuracy among the tested settings.

The improvement observed with a training size of 1000 suggests that model performance benefits from additional task-specific data. Detecting AI-generated text requires recognizing subtle linguistic and stylistic patterns, and larger training datasets expose the model to a wider variety of such examples. In Figure 2, smaller training sizes (50, 100, and 500 samples) produce nearly identical validation accuracy values, indicating limited learning due to insufficient data. Increasing the training size to 1000 provides more representative examples, which improves the model's ability to generalize.

The learning rate of $2e-5$ lies within the commonly recommended fine-tuning range for transformer-based models, typically between $1e-5$ and $5e-5$ (Devlin et al., 2019). Across the tested learning rates ($1e-5$, $2e-5$, and $3e-5$), validation accuracy remained largely unchanged. This suggests that all three values fall within a stable optimization region where the training dynamics are similar. A learning rate that is too large can lead to unstable updates, while a learning rate that is too small may slow convergence or prevent the model from adapting effectively to the task. (Sun et al., 2019).

4 Future Work

A potential improvement to the current implementation include increasing validation size. In Figure 1, the lowest validation accuracy value was about 0.495 and the highest validation accuracy value was about 0.525. This suggests the DistilBERT model may struggle to distinguish human vs AI text, possibly due to the limited validation size of 2000. The training subsets were also much smaller, and the experimental setup deviates from the standard 8:1:1 split, limiting the model's ability to learn robust representations. Another extension would be to test larger training data sizes, which may lead to better generalization.

Another improvement would be to test a larger range of learning rates. The tested learning rates of $1e-5$, $2e-5$, and $3e-5$ for this implementation were close in range. Testing additional values such as $1e-4$ and $1e-6$ may result in a lower or higher validation accuracy values.

Other future extensions include exploring other pretrained transformer models, such as BERT and RoBERTa, which may improve classification accuracy. While DistilBERT offers computational efficiency due to its smaller size, larger models like BERT or RoBERTa typically achieve stronger contextual representations due to deeper architectures and more extensive pretraining. Comparing these models under identical hyperparameter settings would provide insight into the tradeoff between computational cost and detection performance.

References

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